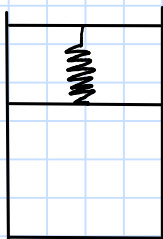


LIBRO CAMILA

$$1\text{ l} = 10^{-3}\text{ m}^3$$

$$1 = 10^{-3}\text{ cm}^3$$

11.9 Area pistone = $200(\text{cm}^2)^{3/2}$
 $V_0 = 5\text{ l}$ $P_0 = 1\text{ atm}$ (india a riposo) $T_0 = -30^\circ\text{C}$
 $T_f = 27^\circ\text{C}$ sollevamento pistone $h = 2\text{ cm}$



$$P_f = ?$$

$$PV = nRT$$

$$V_f = ?$$

$$m = ?$$

$$m = n \cdot M$$

numero di moli
peso molecolare

$$n = \frac{PV}{RT} = \frac{1.5}{82,057 \cdot 10^{-3} \cdot 243} = 0,25\text{ moli}$$

da m^3 a l

$$m = 0,25 \cdot 32 = 8\text{ g}$$

$$V_f = V_0 + \Delta h = 5 + \underbrace{200 \cdot 2 \cdot 10^{-3}}_{\substack{\text{cm}^2 \cdot \text{cm} \\ 600\text{ cm}^3 \\ 6\text{ m}^3}} = 5,4\text{ l}$$

$$P_f V_f = nRT_f \rightarrow P_f = \frac{nRT_f}{V_f} = \frac{0,25 \cdot 82,07 \cdot 10^{-3} \cdot 300}{5,4} = 1139,86 \cdot 10^{-3} = 1,13\text{ atm}$$

$$K = ? \quad -Kx = F_{el}$$

$$F_{el} + P_0 = P_f \rightarrow B - \frac{Kx}{\delta} = P_f$$

$$1,13 = 1 + \frac{K \cdot 2}{200}$$

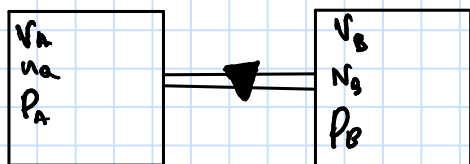
- 2 perché si comprime

$$\frac{100}{200}(1,13 - 1) = K = 13\text{ BOH}$$

$$P = \frac{F}{\delta}$$

$P + P_0$ $\frac{F_{el}}{\delta}$

Utile



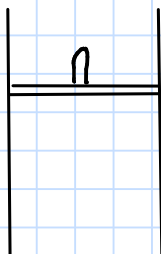
$$V_f = V_A + V_B$$

$$\Delta U = 0 \rightarrow \text{termicamente isolati}$$

$$\Delta U(B) + \Delta U(A) = 0 \rightarrow n_B c_V (T_f - T_A) + n_A c_V (T_f - T_B) = 0 \rightarrow T_f = T_A \quad \text{la } T \text{ non cambia}$$

$$P_f V_f = n R T_f \rightarrow P_f = \frac{n R T_f}{V_f} = \frac{n R T_A}{V_A + V_B}$$

1.7 P_0, T_0
 $V_0 = 10 \text{ l}$

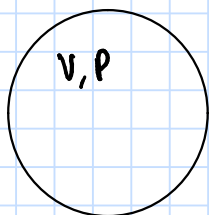


$$V_f = 8 \text{ l} \quad h = ?$$

Isobara $Q - h = \Delta U \quad \Delta U = n c_V dT$

$$h = p_0 dV = \underbrace{1,013 \cdot 10^5}_{1 \text{ atm} = 1 \text{ bar} = 1,013 \cdot 10^5 \frac{\text{N}}{\text{m}^2}} \cdot 2 \cdot 10^{-3} \text{ m}^3 = -202,6 \text{ J} = \text{N} \cdot \text{m}$$

1.8



$$PV = P_0 V_0$$

$$P_0 = 1,5 \text{ atm}$$

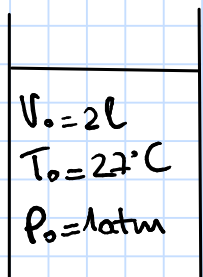
$$V_0 = 20 \text{ l}$$

$$T_0 = \frac{PV}{nR} = \frac{1,5 \cdot 20}{n \cdot 8,314 \cdot 10^{-3}} = \frac{0,36}{n \cdot 10^{-3}} = \frac{360 \text{ K}}{n}$$

$$V_f = 2V_0 = 40 \text{ l} \quad P_f = \frac{1,5 \cdot 20}{40} = 0,75 \text{ atm}$$

$$h = n R T \ln \left(\frac{V_f}{V_0} \right) = \cancel{n} \cdot 8,314 \cdot 10^{-3} \cdot 360 \ln \left(\frac{40}{20} \right) = 29,47 \text{ J}$$

1.10



$$V_f = 2,5 \text{ l}$$

$$n = \frac{PV}{RT} = \frac{1 \cdot 2}{8,314 \cdot 10^{-3} \cdot 300} = \frac{2}{2494,2 \cdot 10^{-3}} = 8,12 \cdot 10^{-2}$$

$$m = n \cdot M = 8,12 \cdot 28 = 227,36$$

$$h = \int_{V_0}^{V_f} p dV = \int_2^{2,5} 1 dV = 1 \text{ atm} (2,5 - 2) = 1 \text{ atm} \cdot 0,5 \text{ l}$$

$$1 \text{ atm} = 1,013 \cdot 10^5 \frac{\text{N}}{\text{m}^2} \quad 1 \text{ l} = 10^{-3} \text{ m}^3$$

$$1,013 \cdot 10^5 \frac{\text{N}}{\text{m}^2} \cdot \frac{1}{2} \cdot 10^{-3} \text{ m}^3 = 50,7 \text{ J}$$

$$Q - h = \Delta U \quad \Delta U = n c_V dT$$

$$\text{isobara} \rightarrow \frac{V_B}{V_A} = \frac{T_B}{T_A} \quad T_B = \frac{V_B}{V_A} \cdot T_A = 375 \text{ K}$$

$$\Delta U = 0,08125 \cdot 1,314 \cdot (375 - 300) = 126$$

$$Q = 126 + 50,7 = 175,41$$

II.21



$$n=2$$

$$T_1 = 270\text{ K}$$

$$T_2 = 280\text{ K}$$

adiabatica reversibile

$$W = p dV = -\Delta U = \Delta U_A + \Delta U_B = 2 \Delta U$$

$$2 n R \ln T = 2 \cdot 2 \cdot 8,314 \cdot \ln 2 = -\frac{498,89\text{ J}}{4,18} = 119,33 \text{ cal}$$

IV.2



$$\eta = 1 - \frac{T_1}{T_2} = 1 - 0,6 = 0,4$$

$$T_c = 520\text{ K} \quad T_f = 300$$

$$\eta = 1 - \frac{300}{520} = 0,42$$

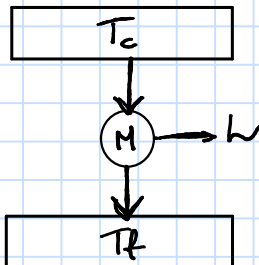
$$T_c = 500\text{ K} \quad T_f = 280$$

$$\eta = 1 - \frac{280}{500} = 0,44$$

IV.15

$$T_c = 500\text{ K} \quad T_f = 300\text{ K} \quad W = 100\text{ J} \quad \eta = 0,20$$

$$\Delta S_u = \Delta S_{\text{calda}} + \Delta S_{\text{fredda}}$$



$$Q - W = \Delta U$$

$$W = Q_{\text{ass}} - Q_{\text{ced}}$$

$$\eta = 1 - \frac{Q_{\text{ced}}}{Q_{\text{ass}}}$$

$$\begin{cases} 100 = Q_{\text{ass}} - Q_{\text{ced}} \\ 0,20 = 1 - \frac{Q_{\text{ced}}}{Q_{\text{ass}}} \end{cases}$$

$$\begin{cases} Q_{\text{ass}} = 100 + Q_{\text{ced}} \\ 0,20 = 1 - \frac{Q_{\text{ced}}}{100 + Q_{\text{ced}}} \end{cases}$$

$$\begin{cases} Q_{\text{ass}} = 100 + Q_{\text{ced}} \\ 0,20 = \frac{100 + Q_{\text{ced}} - Q_{\text{ced}}}{100 + Q_{\text{ced}}} \end{cases}$$

$$\begin{cases} Q_{\text{ass}} = 100 + Q_{\text{ced}} \\ 0,20(100 + Q_{\text{ced}}) = 100 \end{cases}$$

$$\begin{cases} Q_{\text{ass}} = 100 + Q_{\text{ced}} \\ 20 + 0,2Q_{\text{ced}} = 100 \end{cases}$$

$$\begin{cases} Q_{\text{ass}} = 100 + Q_{\text{ced}} = 500 \\ Q_{\text{ced}} = 400 \end{cases}$$

Per una sorgente $\Delta S = \frac{Q_c}{T} = \frac{500}{500} = 1$

$$\Delta S = \frac{Q_a}{T} = \frac{400}{300} = 1,3$$

